

# Technical Progress Report

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|------------------|---------------------|---------------|------------|
| Project name:    | <b>Biped robot</b>  |               |            |
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| Start date:      | 2026-04-20          | End date:     | 2026-04-23 |
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## 1. Executive Summary

This report documents the technical progress of the biped robot project in the current week. The team has completed all necessary preparation work, including gait scheme verification, hardware compartment protection improvement, and vision recognition dataset collection and model training. On this basis, the robot has entered the initial motion attempt stage. During testing, typical issues such as unstable gait, loose joint movement, and occasional shape recognition errors were identified. The team has confirmed the direction of optimization and will carry out targeted improvements to support stable and reliable walking motion in the next stage.

## 2. Objectives & Scope

- Complete the pre-motion preparation including gait design, hardware protection, and vision system debugging.
- Conduct the first attempt of basic walking motion to verify the feasibility of the current structure and control scheme.
- Test the vision recognition function under actual motion conditions and check the recognition effect between cubes and square pyramids.
- Record and analyze problems such as unstable movement and recognition errors during the attempt.
- Lay a foundation for subsequent stable walking and integrated system verification.

## 3. Technical Work Completed

### 3.1 Preparation Work Completion

Finalized and verified the redesigned stepping gait to improve rigidity and support capacity. Completed the inspection and preliminary reinforcement of the hardware compartment to reduce exposed electronic components such as ESP32, camera, battery, and connecting wires. Collected and sorted multi-angle image datasets for shape recognition and finished the retraining of the vision model.

### 3.2 Robot Motion Attempt

Carried out the initial power-on test and basic motion attempt of the biped robot. Verified the joint actuation effect and the coordination between left and right legs during stepping. Recorded the motion trajectory and attitude changes to evaluate the stability of the current mechanical structure.

### 3.3 Vision System Commissioning

Deployed the trained recognition model to the robot system. Tested the real-time performance and accuracy of distinguishing cubes and square pyramids under different angles and distances during the motion attempt.

## 4. Observations & Findings

- The original stepping design is too loose, resulting in weak support and unstable leg movement during walking attempts.
- Excessive components are exposed outside the hardware compartment, which may be damaged if the robot falls during movement.
- The vision system shows confusion between cubes and square pyramids at specific angles.
- Insufficient training data leads to low robustness of the shape recognition model.

## 5.Challenges & Roadblocks

- The residual looseness in the stepping gait restricts the robot from performing stable and continuous walking.
- Incomplete hardware protection limits the intensity and duration of motion tests.
- Limited multi-angle training data leads to unstable recognition results of similar objects at individual viewpoints.
- The coordination between motion control and vision feedback needs further integration and adjustment.

## 6. Next Steps & Recommendations

- Optimize the stepping gait parameters again to enhance support rigidity and eliminate motion looseness.
- Further redesign and fully enclose the hardware compartment to provide reliable protection for internal components.
- Expand the dataset with more diversified angles and scenes, and retrain the vision model to improve recognition robustness.
- Integrate the optimized gait and structure, and conduct continuous and stable walking tests.
- Add real-time vision feedback to the motion control system to improve the overall coordination and reliability.

## 7. Conclusion

This week, the team finished all preparation work and successfully launched the first motion attempt of the biped robot. The basic functions of gait movement, hardware structure, and vision recognition have been verified, but there are still problems including unstable motion, partial exposure of components, and occasional



shape recognition errors. In the next stage, the team will focus on gait optimization, hardware compartment upgrading, dataset expansion, and model retraining, so as to realize stable walking motion and accurate vision recognition of the biped robot.